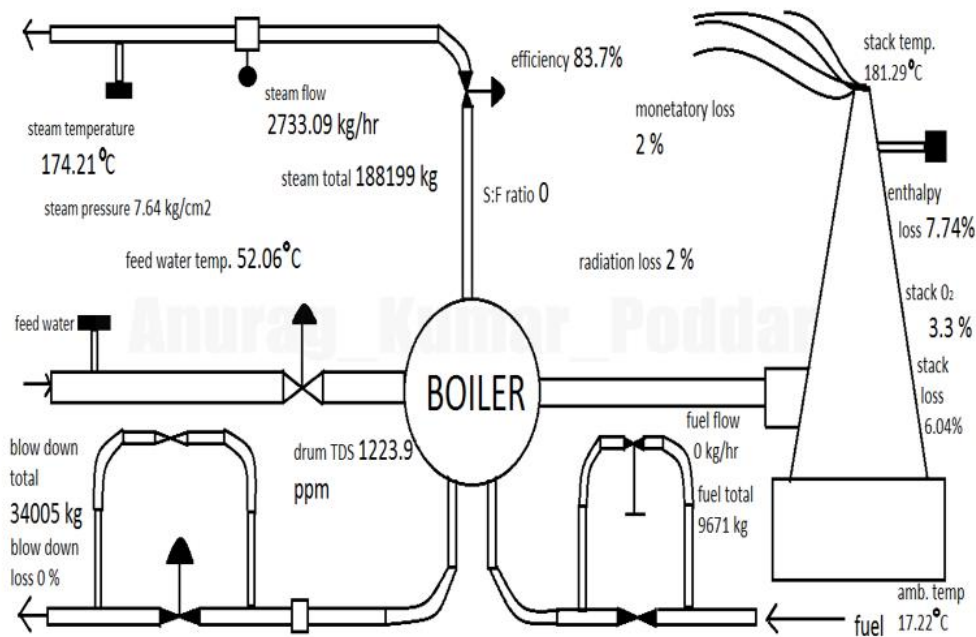


Steam Consumption Optimization in Manufacturing in ITC Project



Project Overview

This project is a comprehensive operational deep-dive conducted within ITC Limited's manufacturing facilities, focusing on steam consumption optimization in the Primary Manufacturing Department (PMD).

It spans the entire value chain — from production processes to resource management — with the objective of improving energy efficiency and operational performance in tobacco manufacturing.

Objectives

- Understand boiler operations and related working principles
- Analyze key processes in the primary manufacturing department
- Quantify:
 - Water added & removed
 - Heat requirements
 - Steam consumption across units

- Establish ideal steam consumption benchmarks
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Process Coverage

The study includes detailed analysis of the following manufacturing units:

- **DCCC (Drying & Conditioning Chamber)**
 - **Lamina Dryer**
 - **Admoisture Unit**
 - **Stem Dryer**
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Key Results

Process Unit	Steam Consumption (kg)
DCCC	241.9 kg
Lamina Dryer	691.05 kg
Admoisture	147.82 kg
Stem Dryer	683.46 kg
Total (PMD)	1764.23 kg

Key Achievements

-  Developed a data-driven model for steam consumption calculation
 -  Identified ideal benchmark consumption across all process units
 -  Highlighted opportunities to reduce energy loss
 -  Improved understanding of industrial energy optimization
 -  Bridged mechanical engineering concepts with data analytics
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Skills & Tools Applied

-  Data Analysis & Interpretation
 -  Industrial Process Analysis
 -  Mechanical Systems Understanding
 -  Efficiency Optimization
 -  Excel
-

Project Documentation

Includes:

- Process analysis
 - Analysis of whole Data
 - Calculations & formulas
 - Engineering assumptions
 - Final conclusions and industrial insights
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Conclusion

This project demonstrates how data-driven analysis can significantly improve industrial efficiency. By calculating and benchmarking steam consumption, the study provides actionable insights to maximize output while minimizing energy losses, making it highly relevant for data analytics in manufacturing and operations.

Acknowledgement

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Let's Connect

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